

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

CELL, The Unit of Life

LECTURE-8

ZOOLOGY

DAVOOD SIR (CEO of NG)



Topics To Be Covered

- 1 Prokaryotic Cell.
- 2 Cell Envelope and its modifications.
- ③ NEET PYQ.

Eukaryotic Cell



Prokaryotic Cell

→ Bacteria
 → Blue Green Algae
 → Mycoplasmas

ancient + Nucleus

Bacteria

Gram Positive → Retains Grams stain.

Gram Negative → Not Retains Grams stain.

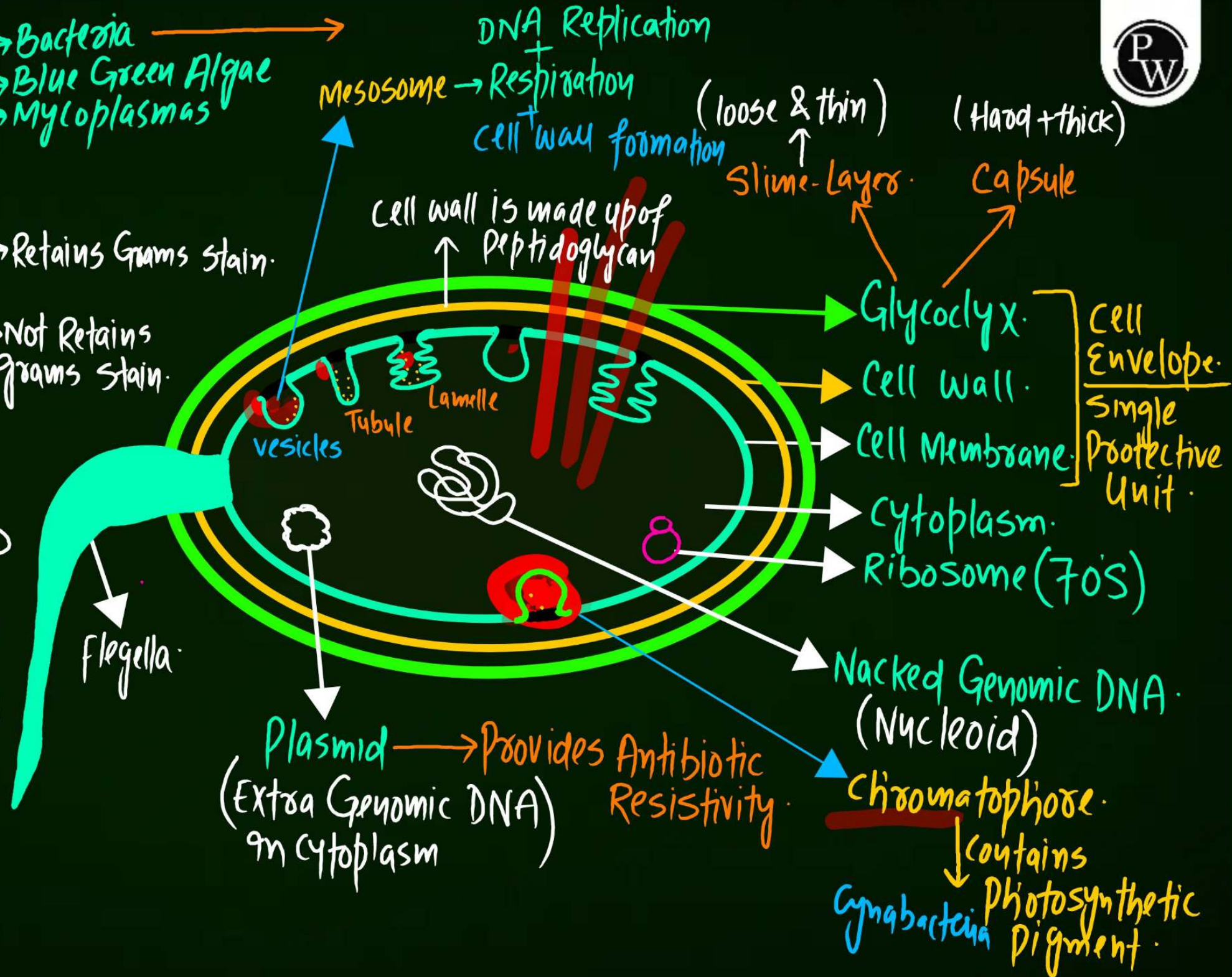
Shape

① COCCUS → Round shape → ○

② Bacillus → Rod-Shaped → ○

③ Vibrio → Comma shaped. ○

④ Spirillum → Spiral shaped. ○



Prokaryotic Cell

The prokaryotic cells are represented by **bacteria, blue-green algae, mycoplasma and PPLO (Pleuro Pneumonia Like Organisms)**. They are generally smaller and multiply more rapidly than the eukaryotic cells. They may vary greatly in shape and size. The four basic shapes of bacteria are:-

a) Bacillus (rod like), b) Coccus (spherical), c) Vibrio (comma shaped), d) Spirillum (spiral).

All prokaryotes have a cell wall surrounding the cell membrane **except in mycoplasma**. The fluid matrix filling the cell is the cytoplasm. There is **no well-defined nucleus (Nucloid)**. The genetic material is not enveloped by a nuclear membrane (**naked**). In addition to the genomic DNA (the single chromosome/circular DNA), many bacteria have small circular DNA outside the genomic DNA called **plasmids**. The plasmid DNA confers certain unique phenotypic characters to such bacteria.

- One such character is **resistance to antibiotics**.
- This plasmid DNA is used to **monitor bacterial transformation with foreign DNA**.

Cell organelles are not found in prokaryotic cells except for **ribosomes (70 S)**. Prokaryotes have something unique in the form of **inclusions**. A specialised differentiated form of cell membrane called **mesosome (infoldings of cell membrane)** is the characteristic of prokaryotes.

Cell Envelope and its Modifications

The bacterial cells, have a chemically complex **cell envelope**. The cell envelope consists of a tightly bound three layered structure i.e.,

- a) The **outermost glycocalyx**,
- b) The middle **cell wall**,
- c) The inner **plasma membrane**.

Although each layer of the envelope performs distinct function, they act together as a **single protective unit**.

Bacteria can be classified into two groups on the basis of the differences in the cell envelopes and the manner in which they respond to the staining procedure developed by **Gram** viz.,

- I. **Gram positive Bacteria:-** those that take up the **gram stain**.
- II. **Gram negative bacteria:-** and that do not take up gram stain.

Cell Envelope and its Modifications

1) Glycocalyx:-

- It Varies in composition and thickness among bacteria & has two forms:
 - a) Slime layer → loose, thin sheath.
 - b) Capsule → thick, tough covering

2) Cell Wall:-

- It determines shape of the bacterial cell & provides strong structural support.
- Prevents bursting or collapsing of the cell.

3) Plasma Membrane:-

- It is selectively permeable & interacts with the external environment.
- It is structurally similar to eukaryotic plasma membrane.
- ✓ a) Mesosomes:- They are special membranous infoldings of plasma membrane & are present as vesicles, tubules, and lamellae. Functions as cell wall formation DNA replication & distribution to daughter cells, Respiration, Secretion processes, Increase surface area of plasma membrane, Increase enzymatic content
- b) Chromatophores:- They are other membranous extensions into cytoplasm in Cyanobacteria which contain photosynthetic pigments.

Question: NEET-2025

Which specialised structure in a prokaryotic cell helps in cell wall formation, DNA replication and respiration?

1 Endoplasmic reticulum

2 Mesosome ✓

3 Chromatophores

4 Cristae



Question: NEET-2023

Given below are two statements.

✓ **Statement I:** In bacteria, the mesosomes are formed by the extensions of plasma membrane. ✓

✓ **Statement II:** The mesosomes, in bacteria, help in DNA replication and cell wall formation. ✓

In the light of the above statements, choose the most appropriate answer from the options given below.

1 Both Statement I and Statement II are correct. ✓

3 Both Statement I and Statement II incorrect.

2 Statement I is correct but Statement II incorrect.

4 Statement I is incorrect but Statement II incorrect.

Question: NEET-2020

H/W



Which of the following statements about inclusion bodies is incorrect?

- 1** They are not bound by any membrane.
- 2** These are involved in ingestion of food particles.
- 3** They lie free in the cytoplasm.
- 4** These represent reserve material in cytoplasm.



Question: NEET-2015

Which of the following structures is not found in a prokaryotic cell?

- 1 Mesosome
- 2 Plasma membrane
- 3 Nuclear envelope
- 4 Ribosome



Thank
You